

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE - 3

**PARTICULATE MATTER
(PART-2)**

POSSIBLE MAINS QUESTION

TOPIC:

GS Paper I : Urbanization- their problems and remedies

GS Paper III : Environmental pollution and degradation

- 1** Examine the health impacts of air pollution in India. Also critically comment on **government's response to control pollution**. (12 Marks)(250 words)

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GOVERNMENT STEPS TO PREVENT PM POLLUTION



AQI	Remark	Color Code	Possible Health Impacts
0-50	Good	Green	Minimal impact
51-100	Satisfactory	Light Green	Minor breathing discomfort to sensitive people
101-200	Moderate	Yellow	Breathing discomfort to the people with lungs, asthma and heart diseases
201-300	Poor	Orange	Breathing discomfort to most people on prolonged exposure
301-400	Very Poor	Red	Respiratory illness on prolonged exposure
401-500	Severe	Dark Red	Affects healthy people and seriously impacts those with existing diseases

1 NATIONAL AIR QUALITY INDEX

- 1 Launched by MoEFCC under Swachh Bharat to be monitored by CPCB.
- 2 'One Number- One Colour-One Description' for the common man to judge the air quality within his vicinity.
- 3 Previously NAMP(National Air Monitoring Program) was followed that monitors 4 pollutants- PM_{10} , $PM_{2.5}$, NO_2 , SO_2 .
- 4 The AQI will consider 8 pollutants- PM_{10} , $PM_{2.5}$, NO_2 , SO_2 , **Carbon Monoxide**, **Ozone**, **Ammonia** and **Lead**.

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AIR POLLUTION: THE GRADES



EMERGENCY

PM 2.5 OVER 300 $\mu\text{g}/\text{m}^3$ OR
PM 10 OVER 500 $\mu\text{g}/\text{m}^3$
FOR 48 HOURS

- Introduce odd-even
- Stop entry of trucks, construction activities
- Set up task force to look into shutting schools

SEVERE

PM 2.5 BETWEEN 121 - 250 $\mu\text{g}/\text{m}^3$, PM 10 BETWEEN 351 - 430 $\mu\text{g}/\text{m}^3$

- Close brick kilns, stone crushers, shut down Badarpur power plant
- Intensify public transport

VERY POOR

PM 2.5 BETWEEN 121 - 250 $\mu\text{g}/\text{m}^3$, PM 10 BETWEEN 351 - 430 $\mu\text{g}/\text{m}^3$

- Stop use of diesel generators
- Increase parking fee by 3-4 times
- Increase bus, Metro services
- Stop use of coal/firewood in hotels, open eateries

MODERATE-POOR

PM 2.5 BETWEEN 61 - 120 $\mu\text{g}/\text{m}^3$, PM 10 BETWEEN 101 - 350 $\mu\text{g}/\text{m}^3$

- Stop garbage burning
- Mechanised sweeping on roads, water sprinkling on unpaved roads
- Enforce SC ban on firecrackers
- Use social media to inform people about pollution

2

GRADED RESPONSE ACTION PLAN

1

MoEFCC has notified the GRAP to combat air pollution in NCR Delhi region.

2

Based on the air quality the grades have been classified as **Emergency**, **Severe**, **Very Poor** and **Moderate poor**.

3

It will be enforced by **Environment Pollution Control Authority (EPCA)**.

4

Under this plan, emergency measures will be automatically enforced in NCR if level of PM2.5 breaches 300 micrograms per cubic metre ($\mu\text{g}/\text{m}^3$) or PM10 levels stay above 500 $\mu\text{g}/\text{m}^3$ for 48 hours.

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HARMFUL EFFECTS OF PARTICULATE MATTER

According to the **National Ambient Air Quality Standards(NAAQS)** of the **Central Pollution Control Board (CPCB)** of India, the 24-hour average for

- **PM10** is **100** microgram/cubic metre
- **PM2.5** is **60** microgram/ cubic metre.



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3 TRADITIONAL CHULHAS



UNNAT CHULHAS AS COUNTER MEASURE

UNNAT CHULHA ABHIYAN



Ministry of New &
Renewable Energy
(MNRE)

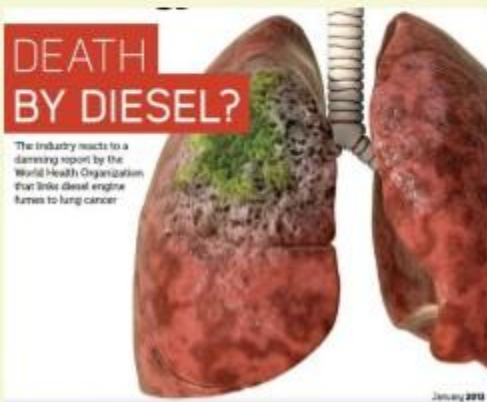
- 1 The Lancet estimated that indoor air pollution-the exhalations of chulhas-causes 400,000 premature deaths every year in India.
- 2 In traditional chulhas, particulate matter concentrations can touch 20,000 micrograms per cubic metre!!

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4 BANNING 10 YEAR OLD DIESEL VEHICLES IN DELHI NCR BY NGT



5 INCREASING USE OF PUBLIC TRANSPORTATION



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6 IMPLEMENTATION OF BS VI

- Ministry of Road Transport & Highways has decided to leapfrog from BS-IV to BS-VI emission norms directly by completely skipping BS-V norms.
- New BS-VI norms will come into effect from 01.04.2020.



7 ODD-EVEN POLICY

- Those with registration numbers ending in odd digits are allowed to ply on odd dates and those ending in even digits run on even dates.
- To discourage private vehicles overcrowding roads and encourage public transportation.



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8

PREVENTION OF STUBBLE BURNING



**CROP RESIDUE
MANAGEMENT**



ALTERNATIVE USES OF PADDY STUBBLE

- 1 Ethanol production
- 2 Electricity:
Punjab alone produces 20 mn tonnes/year of biomass, the full utilization of which will produce 1500MW of electricity as against the current 63 MW.
- 3 **Crop Residue Management**= crop residue incorporation.
- 4 Animal feed



STUBBLE



**POWER
GENERATION**



ANIMAL FEED



**SODIUM
CARBONATE
PRETREATMENT**

FERMENTATION

ETHANOL



FIELD BURNING



National Green Tribunal

**STRICTER IMPLEMENTATION
& PENALTIES BY NGT**

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GOVERNMENT STEPS TO PREVENT PM POLLUTION

9

UTILISATION OF CAMPA FUNDS TO THE MAXIMUM

Compensatory Afforestation Fund Management
and Planning Authority- Lifeline to India's Green
Cover!!



10

GREEN HIGHWAYS= LESS POLLUTION



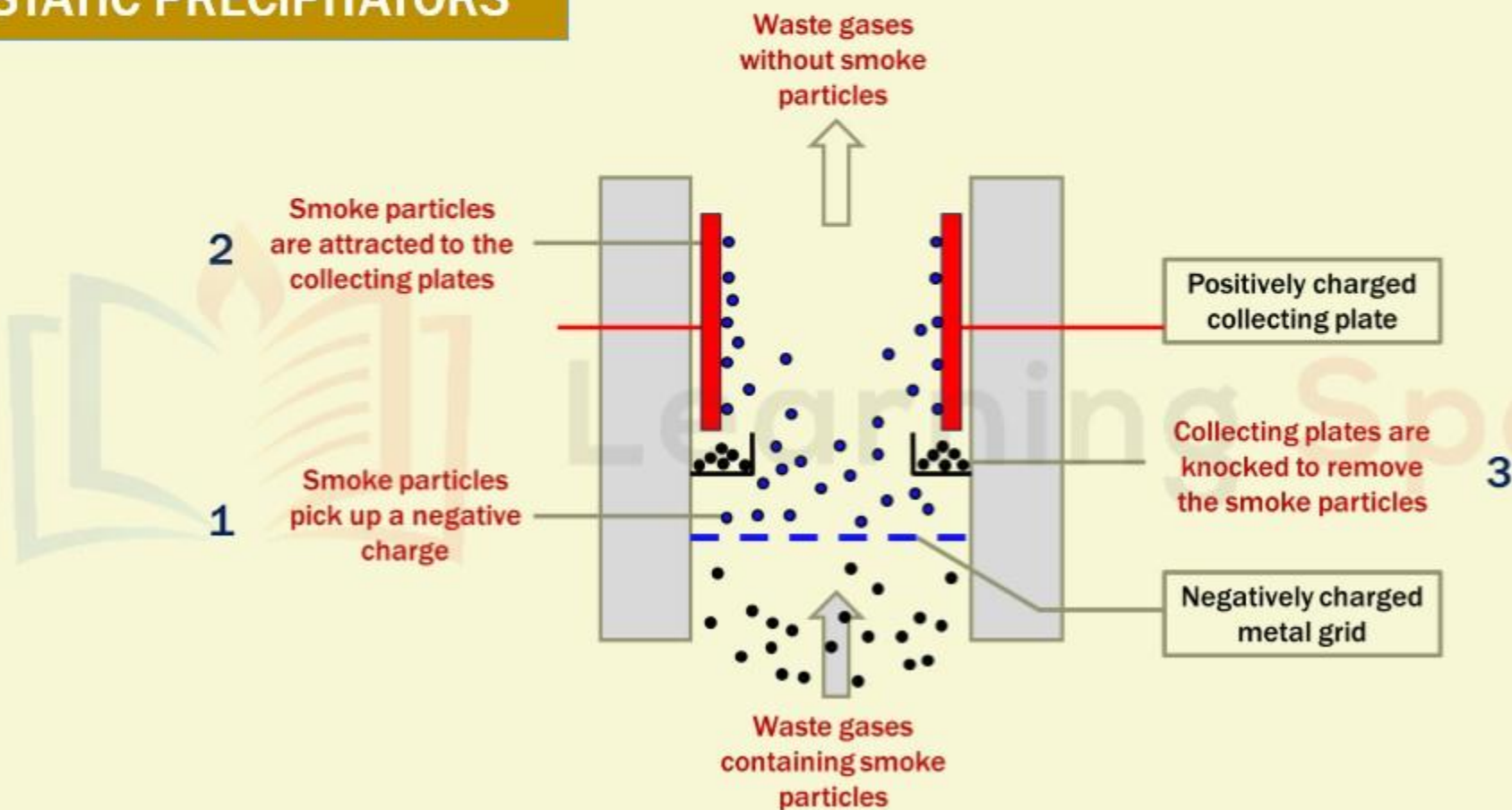
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11 TECHNOLOGICAL SOLUTIONS IN INDUSTRIES AND VEHICLES

ELECTROSTATIC PRECIPITATORS



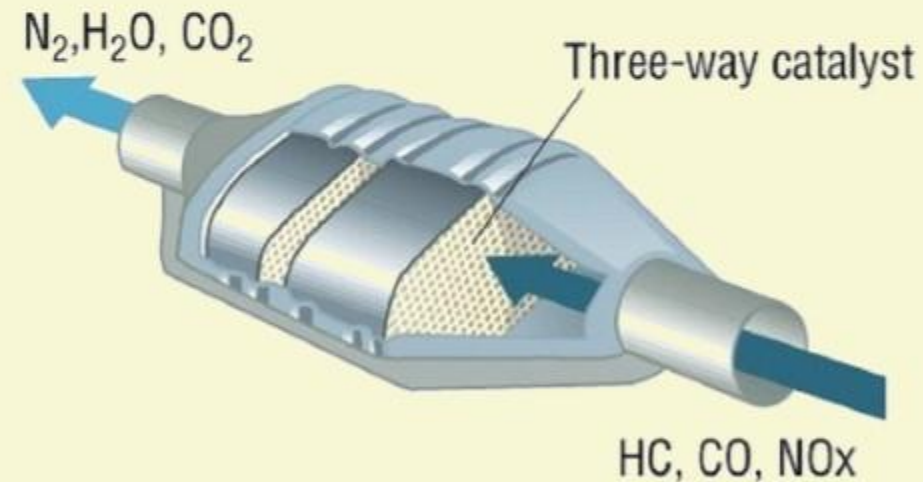
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